

a) $\exists x \forall y (x+y=y)$
 There exists a real number x such that for all real numbers y , the sum $(x+y)$ is equal to y .

Simple: There exists a real number which when added to any other arbitrary real number returns that real number. the set of real numbers exists.
 Or in other words, an additive identity of real numbers exists.

b) $\forall x \forall y ((x > 0) \wedge (y < 0)) \rightarrow (x-y > 0)$

For any two arbitrary real numbers where one is non-negative and the other is negative, the negative real no. when subtracted from the non-negative real no. will return a positive real no.

c) $\exists x \exists y ((x < 0) \wedge (y < 0)) \wedge (x+y > 0)$

There exists two arbitrary non-positive real no.s s.t. their difference is positive.

d) $\forall x \forall y ((x \neq 0) \wedge (y \neq 0)) \leftrightarrow (xy \neq 0)$

For any two arbitrary non-zero real no.s, their product must be non-zero.

25) Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.

a) $\exists x \forall y (xy=y)$
 There exists a real no. s.t. ^{when multiplied with} any arbitrary real number returns that real no.

Or in other words, an multiplicative identity of real no.s exist.

b) $\forall x \forall y (((x < 0) \wedge (y < 0)) \rightarrow (xy > 0))$

For any two arbitrary negative real no.s, their product must be positive.

c) $\exists x \exists y ((x^2 > y) \wedge (x < y))$

There exists a pair of real numbers such that the square of the smaller one is greater than the larger one.